ONLINE STORE GULISTON — AI/AR/ECOSYSTEM TEXNIK TOPSHIRIQ

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Maqsad: Guliston va keyinchalik O‘zbekiston bozoriga mo‘ljallangan lokal e-commerce ekotizimi: marketplace, tezkor logistika, AI marketing, AI chatbot, AR/3D mahsulot ko‘rish, to‘lovlar, xarita, PWA va mobil ilova.

> Muhim xavfsizlik eslatmasi: oldingi handoff hujjatida Telegram bot tokeni ochiq yozilgan. Uni darhol BotFather orqali yangilang (revoke/rotate), eski tokenni ishlatishni to‘xtating va barcha maxfiy kalitlarni faqat `.env`/secret manager’da saqlang. Git tarixidan ham olib tashlang.

1. Vizyon va mahsulot pozitsiyasi

1.1. Muammo

Guliston va Sirdaryo bozorida xaridorlar bir joydan mahsulot qidirish, do‘konni solishtirish, ishonchli yetkazib berish, tez to‘lov va buyurtmani kuzatish imkoniyatiga muhtoj. Mahalliy do‘konlar esa raqamli katalog, marketing dizayni, reklama, mijoz tahlili va logistika boshqaruvida qiynaladi.

1.2. Yechim

Online Store Guliston quyidagi tizimlarni bir platformada birlashtiradi:

- Mahalliy do‘konlar marketplace’i
- Xaridor uchun web/PWA va Flutter mobil ilova
- Shop owner uchun admin kabineti
- Kuryer uchun mobil/PWA kabineti
- AI marketing agenti va dizayn generatori
- AI chatbot: qidirish, savatga qo‘shish, buyurtma, tracking, yordam
- 3D/AR mahsulot ko‘rish va QR-kod
- Xarita, yetkazib berish zonalari va real vaqt tracking
- Click, Payme, Uzum/UzCard/Humo va naqd to‘lovlar
- CRM, sodiqlik, referral, reklama va analitika

1.3. Differensial afzallik

Platforma “eng katta marketplace” bo‘lishdan oldin “eng foydali lokal commerce operating system” bo‘lishi kerak:

1. Mahalliy assortiment va 1–2 soatlik yetkazib berish
2. Do‘konlar uchun past komissiya va kuchli savdo vositalari
3. Uzbek tilida tabiiy ishlaydigan AI yordamchi
4. Ishonch: aniq yetkazish vaqti, real stock, qaytarish qoidasi, verifikatsiya
5. AI poster, katalog kontenti va AR ko‘rish orqali sotuvni oshirish
6. Telegram-first oqimlar: buyurtma, kuryer va promo xabarlari

2. Hozirgi platforma holati

Mavjud stack:

- Django + Django REST Framework + SimpleJWT
- SQLite (MVP), production uchun PostgreSQL
- Django Templates + Tailwind CDN + Alpine.js
- Leaflet + OpenStreetMap + OSRM
- Telegram bildirishnomalari
- UZ/RU/EN til tizimi
- Service Worker va offline-first cache
- Customer, Admin, Courier, Payment, Map modullari

2.1. Productionga majburiy o‘tishlar

- SQLite → PostgreSQL 16+
- Local media → S3-compatible storage (Cloudflare R2, AWS S3 yoki MinIO)
- Redis: cache, Celery broker, chatbot session, rate-limit
- Celery + Celery Beat: background tasks va retry
- Nginx + Gunicorn/Uvicorn
- Docker Compose va CI/CD
- Sentry monitoring
- `.env` va secret manager
- `DEBUG=False`, HSTS, CSRF/CORS, security headers
- Avtomatik backup: DB har kuni, media har hafta

3. Bozor va raqobat strategiyasi

O‘zbekistonda Uzum, Olcha, Asaxiy, Wildberries va tezkor delivery servislar xaridor e’tiborini egallaydi. Katta platformalar assortiment, nasiya va brend kuchi bilan ustun; lokal platformaning imkoniyati esa yaqin do‘kon, tez xizmat, real stock, mahalliy til, individual servis va past savdogar komissiyasidadir.

3.1. Raqobat jadvali

Mezoni	Katta marketplace	Tez delivery servisi	Online Store Guliston maqsadi
Assortiment	Juda keng	Oziq-ovqatga yaqin	Lokal ehtiyojlar + vertikal kategoriyalar
Yetkazish	1–bir necha kun	20–60 daqiqa	30–120 daqiqa, zona bo‘yicha
Mahalliy do‘kon	Komissiya va qoidalar qattiq	Cheklangan	Past komissiya, oson onboarding
AI marketing	Cheklangan yoki yopiq	Cheklangan	Pro seller uchun avtomatik poster/kontent
AR/3D	Noyob	Deyarli yo‘q	Mebel, kiyim, tort/pitsa, premium mahsulotlar
Support	Mass-market	Mass-market	Telegram/telefon/live chat, lokal operator
Xarita	Cheklangan	Kuchli	Do‘kon + kuryer + pickup + delivery zone

3.2. 5 yilga ustunlik beradigan prinsiplar

- Avval bitta shaharni ideal qiling; keyin viloyatga kengaying.
- “Hamma mahsulot” emas, eng tez-tez sotiladigan 5–8 kategoriya bilan product-market fit toping.
- Har bir buyurtma bo‘yicha SLA o‘lchang: qabul qilish, tayyorlash, olib ketish, yetkazish vaqti.
- Marketplace emas, seller SaaS ham soting: AI poster, CRM, analitika, reklama va stock yordamchisi.
- AI va AR’ni reklama emas, konversiya o‘lchovi bilan joriy qiling.
- Yandex yoki boshqa agregatorga to‘liq qaram bo‘lmang; gibrid logistika qiling.

4. Foydalanuvchi rollari

4.1. Xaridor

- Ro‘yxatdan o‘tish: telefon OTP, Telegram yoki email
- Katalog, qidiruv, filtrlash, savat
- Shop va mahsulot baholash
- Xarita orqali yaqin do‘konlar
- Yetkazib berish yoki pickup
- To‘lov, tracking, qayta buyurtma

- AI chatbot va voice input
- AR/3D view, QR scan
- Bonus, referral, promokod

4.2. Do‘kon egasi

- Do‘kon profili, ish vaqti, zonalar, minimal buyurtma
- Mahsulot CRUD, ombor qoldig‘i, narx, aksiyalar
- Pro tarifda AI dizayn va marketing agent
- Buyurtma qabul qilish va tayyorlash statusi
- Sotuv, konversiya, top mahsulotlar analitikasi
- Reklama kampaniyasi va sponsor slotlari
- Xodim rollari: owner, manager, operator, packer

4.3. Kuryer

- Login va online/offline status
- Buyurtmani olish/rad etish
- Pickup va delivery navigatsiyasi
- Jonli lokatsiya yuborish
- OTP yoki tasdiqlash kodi bilan topshirish
- Kunlik daromad va performance

4.4. Platforma admini

- Seller verifikatsiyasi
- Kategoriya, komissiya, delivery zone, promo boshqaruvi
- Moderatsiya: mahsulot, rasm, AI natija, review
- To‘lov reconciliation
- Kuryer va logistika monitoringi
- AI xarajat limiti va audit
- Fraud/risk monitoring

5. Mahsulot modullari

5.1. Core commerce

- Katalog, kategoriya, brend, tag
- Full-text qidiruv va synonymlar
- Variantlar: rang, o‘lcham, vazn, hajm
- Inventory reservation
- Multi-shop cart: MVPda do‘konlar bo‘yicha alohida checkout; keyin splitting
- Promo code, bundle, upsell, cross-sell
- Wishlist va recently viewed
- Review, foto-review, Q&A

5.2. Delivery va logistics

- Delivery zonasi: poligon yoki radius
- Dinamik delivery fee: masofa, vaqt, og‘irlik, talab
- ETA: do‘kon tayyorlash va kuryer masofasiga bog‘liq
- Courier assignment: auto-dispatch + manual override
- Pickup point
- Delivery proof: OTP, foto yoki signature
- Failed delivery va refund workflow

5.3. Marketing va CRM

- Segmentlar: yangi, faol, qaytmayotgan, VIP
- Trigger kampaniyalar: abandoned cart, reorder, birthday, price drop
- Referral: taklif qiluvchi va yangi foydalanuvchiga mukofot
- Loyalty: ball, level, cashback
- A/B testing: banner, promo, delivery threshold
- Consent management: marketing xabarlari uchun ruxsat

5.4. In-app browser

Mobil ilovada mini-brauzer faqat xavfsiz whitelisted URL’lar uchun ishlaydi:

- To‘lov provayderi checkout sahifasi
- Seller verified external catalog yoki video
- Help/FAQ sahifalari
- Deep-link handling

Cheklovlar:

- Foydalanuvchi paroli yoki karta ma’lumotini ilova ichida saqlamaslik

- Faqat HTTPS
- External URL warning
- XSS va redirect allowlist
- Payment callbackni server-side tasdiqlash

6. AI arxitekturasi

6.1. Asosiy qoida

AI har bir so‘rovga chaqirilmasin. Avval deterministic qoidalar, FAQ, qidiruv, intent router va cache ishlasin; faqat murakkab yoki generativ vazifada model ishlatilsin. Bu token xarajatini ham, javob vaqtini ham kamaytiradi.

6.2. AI qatlamlari

1. Rule/FAQ layer: 0 token
2. Intent router: kichik model yoki klassifikator
3. Tool execution: database/API operatsiyalari
4. Response formatter: templated lokal javob
5. LLM fallback: murakkab savol, mahsulot tavsiyasi, marketing kontent
6. Human handoff: operatorga eskalatsiya

6.3. Tavsiya etilgan provayder strategiyasi

Bir provayderga qattiq bog‘lanib qolmaslik uchun adapter qatlam yoziladi:

- **Primary text/reasoning:** OpenAI’ning kichik/tejamkor modeli yoki ekvivalent model
- **Fallback text:** boshqa provayderdagi arzon va tez model
- **Embeddings:** multilingual embedding provider
- **Image generation:** OpenAI image API yoki boshqa tijoriy image API
- **Image moderation:** provider moderation + server rules
- **Speech-to-text:** browser Web Speech API MVP; productionda server-side speech provider

Model nomi va narxi tez o‘zgaradi; konfiguratsiya `AI_PROVIDER`, `AI_TEXT_MODEL_FAST`, `AI_TEXT_MODEL_QUALITY`, `AI_IMAGE_MODEL` environment variable orqali boshqarilishi shart. OpenAI narxlari va limitlari doimiy yangilanadi, shuning uchun productiondan avval rasmiy pricing sahifasi tekshiriladi. Batch API mavjud bo‘lsa, non-real-time ishlar uchun ishlatiladi.

6.4. AI xizmatlari

Xizmat	Input	Output	Token nazorati
Product copywriter	Mahsulot atributlari	nom, tavsif, SEO, tag	JSON, max 400 output token
Poster generator	mahsulot rasmi + brief	banner/poster	Pro kredit, moderation
Translation assistant	source text	UZ/RU/EN	cache by content hash
Chatbot	user message + minimal context	text + tool action	intent-first, RAG, cache
Seller copilot	sales/order data	insight/action	daily batch report
Search query parser	natural language	filter JSON	max 100 token
Review summarizer	reviews	qisqa insight	async batch

7. AI dizayn generatori

7.1. Pro tariflar

Tarif	Narx modeli	AI poster	AI text	AR model	Ranking imtiyozi
Free	bepul	0–cheklangan	cheklangan	yo‘q	yo‘q
Pro	oylik + kredit	oylik kredit	yuqori limit	cheklangan	sifat signali
Business	oylik	ko‘p kredit	ko‘p tilli paket	ko‘proq	dashboard/reklama

Muhim: Pro mahsulotni avtomatik birinchi o‘ringa chiqarish adolatsiz va ishonchni kamaytirishi mumkin. “Sponsored” natijalar albatta **Reklama** deb belgilanishi, organik ranking esa sifat, mavjud stock, narx, reyting, yetkazish va relevancelarga asoslanishi kerak.

7.2. Dizayn oqimi

1. Seller mahsulotni tanlaydi.
2. Brief beradi: aksiya, narx, maqsad platforma, til, format.
3. Tizim brend ranglari, logo, mahsulot rasmi va qonuniy cheklovlarni oladi.
4. Prompt builder qisqa structured prompt yaratadi.
5. Image provider 1–3 variant yaratadi.
6. Moderation va text overlay tekshiruvi bajariladi.
7. Seller edit qiladi: narx, CTA, rang, format.
8. Natija media storagega saqlanadi.
9. Instagram/Telegram uchun eksport yoki scheduling taklif qilinadi.

7.3. Prompt shabloni

```
Create a clean commercial poster for an Uzbek local e-commerce marketplace.
Product: {product_name}.
Product category: {category}.
Brand colors: {brand_colors}.
Offer: {price} UZS, {discount}% discount.
Audience: {audience}.
Layout: product hero centered, clean space for Uzbek headline, mobile-first 1080x135
Style: photorealistic, premium, minimal, high contrast, no illegible text, no waterma
Do not invent claims, certifications, ingredients, or prices.
```

7.4. AI dizayn uchun model maydonlari

```
class AIGeneration(models.Model):
    STATUS = [
        ('queued', 'Queued'), ('processing', 'Processing'),
        ('completed', 'Completed'), ('failed', 'Failed'), ('rejected', 'Rejected'),
    ]
    shop = models.ForeignKey('shops.Shop', on_delete=models.CASCADE)
    product = models.ForeignKey('products.Product', null=True, blank=True, on_delete=
    kind = models.CharField(max_length=30)  # poster, copy, translation, ar
    input_payload = models.JSONField(default=dict)
    output_payload = models.JSONField(default=dict)
    provider = models.CharField(max_length=50)
    model = models.CharField(max_length=100)
    status = models.CharField(max_length=20, choices=STATUS, default='queued')
    input_tokens = models.IntegerField(default=0)
    output_tokens = models.IntegerField(default=0)
    estimated_cost_uzs = models.DecimalField(max_digits=12, decimal_places=2, defaul
    error_message = models.TextField(blank=True)
    created_at = models.DateTimeField(auto_now_add=True)
    completed_at = models.DateTimeField(null=True, blank=True)
```

7.5. Token va xarajat cheklovlari

- Har bir seller uchun kunlik va oylik kredit.
- Har so‘rovda max input/output token.
- Cache key: `hash(product_id + locale + template + inputs)`.
- Bir xil poster request qayta yuborilsa cache preview qaytariladi.
- Rasm generatsiyasi async Celery task orqali.
- Failed task uchun exponential backoff va maximum retry.
- Admin dashboardda provider, model, token, UZS xarajat ko‘rinsin.

8. AI chatbot va agent

8.1. Chatbot vazifalari

Xaridor:

- “5 ming so‘mgacha non top” kabi tabiiy til qidiruvi
- Mahsulot taqqoslash
- Savatga qo‘shish (tasdiq bilan)
- Buyurtma statusi
- Yetkazish narxi va ETA
- Promo/bonus izlash
- Qayta buyurtma
- Operatorga ulash

Seller:

- “Bugun nechta buyurtma bo‘ldi?”
- “Sut mahsulotlariga 10% aksiya yarat”
- “Mahsulot tavsifini rus tiliga tarjima qil”
- “Poster yarat”

Courier:

- Bugungi topshiriqlar
- Marshrutni ochish
- Statusni yangilash

Admin:

- KPI so‘rash
- Moderatsiya queue ko‘rish
- Hisobot eksportiga so‘rov

8.2. Agent xavfsizlik qoidasi

Chatbot ilovani “to‘liq boshqarishi” mumkin, ammo har qanday state-changing action uchun quyidagi tartib majburiy:

1. User intent aniqlanadi.
2. Role va permission tekshiriladi.
3. Tool faqat allowlistdan tanlanadi.
4. Action preview ko‘rsatiladi.
5. Foydalanuvchi explicit confirm qiladi.

6. Idempotency key bilan amal bajariladi.
7. Audit log yoziladi.

Misollar:

- Savatga qo‘shish: tasdiqlash talab qilinmaydi yoki settingga bog‘liq.
- Buyurtma yaratish: manzil, summa, to‘lov usuli tasdiqlanadi.
- Mahsulot narxini o‘zgartirish: seller confirm qiladi.
- Refund, payout, delete: qat’iy permission + confirm + audit.

8.3. Tool schema namunasi

```
{
  "name": "search_products",
  "description": "Katalogdan mahsulot qidiradi",
  "parameters": {
    "type": "object",
    "properties": {
      "query": {"type": "string"},
      "category_id": {"type": "integer"},
      "max_price": {"type": "integer"},
      "latitude": {"type": "number"},
      "longitude": {"type": "number"},
      "language": {"type": "string", "enum": ["uz", "ru", "en"]}
    },
    "required": ["query", "language"]
  }
}
```

8.4. Intentlar

```
product_search
product_details
add_to_cart
cart_view
checkout_start
order_tracking
repeat_order
payment_help
delivery_help
promo_help
review_create
seller_product_create
seller_product_update
seller_ai_poster
seller_analytics
courier_tasks
operator_handoff
general_faq
```

8.5. RAG va knowledge base

- FAQ, delivery policy, refund policy, seller guide va category information markdown hujjatlardan indexlanadi.
- Har til uchun alohida yoki til-aware chunk metadata.
- Qidiruv: avval keyword/BM25, keyin embedding hybrid search.
- RAG javobi manbaga bog‘lanadi; noma’lum ma’lumot uydirilmaydi.
- Mahsulot narxi/stock hech qachon RAG’dan emas, to‘g‘ridan-to‘g‘ri DB/API’dan olinadi.

8.6. Chat session ma’lumoti

```
{
  "session_id": "uuid",
  "user_id": 123,
  "locale": "uz",
  "role": "customer",
  "last_intent": "product_search",
  "cart_id": 45,
  "selected_shop_id": 10,
  "recent_product_ids": [1, 2, 3],
  "summary": "User seeks bread under 5000 UZS near location.",
  "updated_at": "ISO-8601"
}
```

Har 8–12 xabardan keyin suhbat summaryga siqiladi. To‘liq chat tarixi ehtiyoj bo‘lsa DBda saqlanadi, ammo modelga faqat oxirgi muhim context yuboriladi.

9. Qidiruv va ranking

9.1. Organik ranking

Qidiruv natijasi quyidagilar bilan hisoblanadi:

- Query relevance
- Stock availability
- Narx raqobatbardoshligi
- Reyting va verified review
- Buyurtma bajarish sifati
- Yetkazish ETA va masofa
- Mahsulot kontenti sifati: rasm, tavsif, atribut, video/3D
- Seller cancellation rate
- Freshness va aksiya

9.2. Sponsored natijalar

- Natija tepasida maksimum 1–3 “Reklama” belgili kartalar.
- Reklama organik natijadan vizual ajratiladi.
- Past rating, stock yo‘q yoki SLA yomon seller reklama qilolmaydi.
- Click/impression/conversion va budjet tracking bo‘ladi.

9.3. Feature scoring namunasi

```
score = (
    relevance * 0.35 +
    availability * 0.15 +
    service_quality * 0.15 +
    rating_quality * 0.10 +
    delivery_score * 0.10 +
    content_quality * 0.10 +
    promotion_score * 0.05
)
```

Bu formula A/B test bilan o‘zgartiriladi. Pro bo‘lish kontent vositasi va reklama paketini beradi, lekin organik qidiruvni yashirin manipulyatsiya qilmaydi.

10. AR/3D tizimi

10.1. Qaysi kategoriya uchun

AR hamma mahsulotga bir xil foyda bermaydi. Birinchi navbatda:

- Mebel: stol, stul, shkaf
- Uy jihozlari
- Kiyim/aksessuar: avval 3D preview, keyin virtual try-on
- Tort, pitsa va premium ovqat: 3D showcase
- Elektronika: o‘lchamni ko‘rsatish
- Sovg‘a va dekor

Meva, oddiy paketli mahsulot va kundalik FMCG uchun ko‘pincha sifatli foto/video, aniq og‘irlik va real review AR’dan foydaliroq bo‘ladi.

10.2. Pipeline

1. Seller 40–120 ta foto yoki 360 video yuklaydi.
2. Upload validation: format, blur, minimum burchak, fon.
3. Photogrammetry/3D provider job yaratiladi.
4. Background worker job statusini polling/webhook bilan kuzatadi.

5. GLB va iOS uchun USDZ asset olinadi.
6. Blender/mesh pipeline bilan polygon, texture va scale optimizatsiya qilinadi.
7. Physical dimensions metadata orqali metrga mos scale o‘rnatiladi.
8. CDNga joylanadi.
9. AR URL va QR code yaratiladi.
10. QA: Android Chrome, Android Scene Viewer, iOS Quick Look, desktop 3D preview.

10.3. Formatlar

- `.glb`: web va Android uchun asosiy format
- `.usdz`: iOS Quick Look uchun
- Texture: WebP/KTX2 imkon bo‘lsa
- Mesh compression: Draco yoki Meshopt
- LOD: low/medium/high

`<model-viewer>` web sahifada interaktiv 3D ko‘rsatish va WebXR, Android Scene Viewer hamda iOS Quick Look rejimlariga fallback berishi mumkin. Rasmiy dokumentatsiya WebXR, Scene Viewer va Quick Look rejimlarini qo‘llab-quvvatlashini ko‘rsatadi. [web:31][web:39]

10.4. Product model kengaytmasi

```
class Product(models.Model):
    # existing fields...
    width_mm = models.PositiveIntegerField(null=True, blank=True)
    height_mm = models.PositiveIntegerField(null=True, blank=True)
    depth_mm = models.PositiveIntegerField(null=True, blank=True)
    weight_g = models.PositiveIntegerField(null=True, blank=True)
    volume_ml = models.PositiveIntegerField(null=True, blank=True)
    glb_file = models.FileField(upload_to='ar/glb/', null=True, blank=True)
    usdz_file = models.FileField(upload_to='ar/usdz/', null=True, blank=True)
    ar_qr_code = models.ImageField(upload_to='ar/qr/', null=True, blank=True)
    ar_status = models.CharField(max_length=20, default='none')
    ar_scale_verified = models.BooleanField(default=False)
```

10.5. AR sahifasi

```
<model-viewer
  src="{{ product.glb_file.url }}"
  ios-src="{{ product.usdz_file.url }}"
  alt="{{ product.name }} 3D modeli"
  ar
  ar-modes="webxr scene-viewer quick-look"
  ar-scale="fixed"
  camera-controls
  auto-rotate
  shadow-intensity="1">
```

```
  <button slot="ar-button" class="btn btn-primary">AR da ko‘rish</button>
</model-viewer>
```

10.6. AR sifat qoidalari

- “1:1” deb yozishdan avval haqiqiy o‘lcham verifikatsiya qilinadi.
- Food mahsulotlarda hajm ko‘rsatkichi taxminiy ekanini bildirish mumkin.
- Foydalanuvchi qurilmasi AR’ni qo‘llamasa 3D viewer fallback ishlaydi.
- 3D model 10–20 MBdan oshmasligi, mobil tarmoqqa mos lazy-load bo‘lishi kerak.

11. To‘lov tizimi

11.1. Hozir API kalitlari yo‘q holatda tayyorgarlik

API kalitsiz ham to‘lov modulini “provider adapter” arxitekturasi bilan to‘liq tayyorlab qo‘yish mumkin. Test/mock provider ishlatiladi; real kalit kelganda environment variable orqali provider yoqiladi.

11.2. Provider interface

```
class PaymentProvider:
    code: str

    def create_payment(self, order, return_url, callback_url) -> dict:
        raise NotImplementedError

    def verify_callback(self, request) -> dict:
        raise NotImplementedError

    def get_status(self, transaction_id: str) -> dict:
        raise NotImplementedError

    def refund(self, payment, amount) -> dict:
        raise NotImplementedError
```

Implementatsiyalar:

```
payments/providers/mock.py
payments/providers/click.py
payments/providers/payme.py
payments/providers/uzum.py
payments/providers/cash.py
```

11.3. Payment modeli

```
class Payment(models.Model):
    STATUS = [
        ('created', 'Created'), ('pending', 'Pending'), ('paid', 'Paid'),
        ('failed', 'Failed'), ('cancelled', 'Cancelled'), ('refunded', 'Refunded'),
        ('partially_refunded', 'Partially Refunded'),
    ]
    order = models.ForeignKey('orders.Order', on_delete=models.PROTECT, related_name=
    provider = models.CharField(max_length=30)
    status = models.CharField(max_length=30, choices=STATUS, default='created')
    amount = models.DecimalField(max_digits=14, decimal_places=2)
    currency = models.CharField(max_length=3, default='UZS')
    merchant_transaction_id = models.CharField(max_length=100, unique=True)
    provider_transaction_id = models.CharField(max_length=100, blank=True)
    payment_url = models.URLField(blank=True)
    raw_request = models.JSONField(default=dict)
    raw_response = models.JSONField(default=dict)
    paid_at = models.DateTimeField(null=True, blank=True)
    created_at = models.DateTimeField(auto_now_add=True)
```

11.4. Payment flow

1. Order serverda yaratiladi: `payment_status=pending`.
2. Tizim idempotent `merchant_transaction_id` yaratadi.
3. Payment provider invoice/link yaratadi.
4. User checkout sahifasiga redirect qilinadi.
5. Provider callback yuboradi.
6. Server signature/authenticationni tekshiradi.
7. Transaction amount va orderni tekshiradi.
8. Callbackni idempotent qayta ishlaydi.
9. Payment paid bo‘lsa order confirmed qilinadi.
10. Seller, customer, courier uchun xabar yuboriladi.
11. Reconciliation job provider statusini periodik tekshiradi.

11.5. Click integratsiyasi

CLICK merchant dokumentatsiyasida merchant API endpointi va invoice yaratish endpointi ko‘rsatilgan; real request formati, signature va callback qoidalari aynan merchant kabineti/dokumentatsiyasidagi amaldagi versiyaga mos yozilishi kerak. [web:32][web:37]

11.6. Payme integratsiyasi

- Merchant account ochiladi.
- Test credentials olinadi.
- JSON-RPC metodlari provider hujjatiga mos yoziladi.
- `CheckPerformTransaction`, `CreateTransaction`, `PerformTransaction`, `CancelTransaction`, `CheckTransaction` oqimlari adapterda ajratiladi.
- `account.order_id` bo‘yicha order topiladi.
- Miqdor tiyin/so‘m birligi aniq test qilinadi.

11.7. Xavfsizlik

- API key hech qachon frontendga chiqmaydi.
- Payment callback IP/signature verification.
- `merchant_transaction_id` unique.
- Callback replayga qarshi idempotency.
- Summa faqat serverdagi orderdan olinadi.
- Karta ma’lumoti platformada saqlanmaydi.
- Har payment uchun audit log.
- Refund faqat role/approval orqali.

11.8. Mock provider

```
class MockPaymentProvider(PaymentProvider):
    code = 'mock'

    def create_payment(self, order, return_url, callback_url):
        return {
            'provider_transaction_id': f'MOCK-{order.order_number}',
            'payment_url': f'/payment/mock/{order.id}/',
            'status': 'pending',
        }
```

Bu orqali UI, order state machine, webhook, success/failure va testlar API kalitsiz tugallanadi.

12. Yetkazib berish: Yandex yoki o‘z kuryeri?

12.1. Tavsiya: gibrid model

Boshlanishda faqat Yandexga tayanish tavsiya etilmaydi. Eng yaxshi model:

- **O‘z kuryeri:** markaziy zonalar, takroriy buyurtmalar, brend nazorati, arzon delivery
- **Hamkor/agregator:** pik soat, uzoq zona, kuryer yetishmovchiligi, shaharlararo jo‘natma

- **Pickup:** do‘kondan olib ketish, delivery xarajatini kamaytirish

Yandex Go O‘zbekistonda shaharlararo jo‘natma xizmatlarini ham kengaytirganini e’lon qilgan, ammo sizning hududingizdagi real API/coverage, tarif va SLA’ni tijoriy vakil bilan tekshirish zarur. [web:14]

12.2. Nega gibrid model

Holat	O‘z kuryeri	Agregator/Yandex	Tanlov
0–3 km, buyurtma ko‘p	Juda foydali	Nisbatan qimmat	O‘z kuryeri
Pik soat	Cheklangan quvvat	Tez scale	Agregator
5+ km	Xarajat oshadi	Narx aniqroq bo‘lishi mumkin	Dinamik tanlov
Brend tajribasi	Yuqori nazorat	Past nazorat	O‘z kuryeri
Yangi shahar	Kuryer yollash qiyin	Tez ishga tushish	Agregator

12.3. Delivery price modeli

```
base_fee + (distance_km × per_km_fee) + peak_fee + weight_fee - free_delivery_discou
```

Misol policy:

- 0–2 km: 5 000 so‘m
- 2–5 km: 7 000 so‘m
- 5–8 km: 10 000 so‘m
- Minimal buyurtma: kategoriya yoki shopga bog‘liq
- Free delivery threshold: do‘kon marjasi bo‘yicha
- Pik vaqtda aniq surcharge ko‘rsatilsin

12.4. Dispatch algoritmi

1. Order ready bo‘lganda courier pool olinadi.
2. Online, mos zona, yuk sig‘imi, hozirgi masofa va acceptance rate score qilinadi.
3. Eng yaxshi 1–3 kuryerga offer yuboriladi.
4. Timeoutdan so‘ng keyingi kuryerga o‘tadi.
5. Manual dispatcher override qila oladi.
6. ETA real GPSga qarab qayta hisoblanadi.

12.5. Delivery KPI

- Acceptance rate
- Pickup on-time rate
- Delivery on-time rate
- Avg delivery minutes
- Cancellation rate
- Failed delivery rate
- Cost per delivery
- Courier earnings/hour
- NPS/CSAT

13. Xarita va navigatsiya

13.1. Mavjud stack

Leaflet + OpenStreetMap + OSRM MVP uchun mos. Productionda tile usage policy, rate limits va geocoding provider alohida ko‘riladi.

13.2. Funksiyalar

- User geolocation consent
- Shop markers
- Delivery zone polygons
- Courier realtime marker
- Order route
- Pickup point
- Address autocomplete/geocoding
- Saved addresses: uy, ish, boshqa
- Entrance/floor/apartment va delivery note

13.3. Lokatsiya maxfiyligi

- Faqat delivery davomida kuryer lokatsiyasi ko‘rsatiladi.
- Customer exact koordinatasi faqat buyurtmaga tayinlangan kuryerga beriladi.
- Lokatsiya retention muddati belgilanadi.
- GPS update interval adaptive: harakatda 10–20 s, fonda kamroq.

14. Ko‘p tillilik: UZ/RU/EN

14.1. Arxitektura

- UI: Django i18n yoki markazlashgan translation catalog
- API: `Accept-Language` hamda `lang` parametri
- DB: mahsulot nomi/tavsifining tarjima JSON maydonlari yoki translation model
- AI: user localega mos output, keyin moderation
- SEO: `/uz/`, `/ru/`, `/en/` yoki hreflang strategy

14.2. Translation model

```
class ProductTranslation(models.Model):
    product = models.ForeignKey('products.Product', on_delete=models.CASCADE, relate
    language = models.CharField(max_length=5, choices=[('uz', 'Uzbek'), ('ru', 'Russ
    name = models.CharField(max_length=200)
    description = models.TextField(blank=True)
    seo_title = models.CharField(max_length=255, blank=True)
    seo_description = models.CharField(max_length=300, blank=True)

    class Meta:
        unique_together = ('product', 'language')
```

14.3. Sifat qoidalari

- Narx, o‘lchov va birliklar tarjimada buzilmasin.
- “dona”, “kg”, “litr” locale formatga mos bo‘lsin.
- Seller kiritgan matn AI tarjimasidan oldin moderationdan o‘tsin.
- Muhim yuridik/to‘lov matnlari inson tomonidan tasdiqlangan translationsdan chiqsin.

15. Ma’lumotlar modeli kengaytmasi

15.1. Shop pro va quality

```
class Shop(models.Model):
    # existing fields...
    plan = models.CharField(max_length=20, default='free')
    plan_expires_at = models.DateTimeField(null=True, blank=True)
    commission_rate = models.DecimalField(max_digits=5, decimal_places=2, default=8.0
    acceptance_rate = models.DecimalField(max_digits=5, decimal_places=2, default=100
    cancellation_rate = models.DecimalField(max_digits=5, decimal_places=2, default=0
    avg_preparation_minutes = models.PositiveIntegerField(default=30)
    is_verified = models.BooleanField(default=False)
```

15.2. Product sifat

```
class Product(models.Model):
    # existing fields...
    sku = models.CharField(max_length=80, blank=True)
    barcode = models.CharField(max_length=80, blank=True)
    description = models.TextField(blank=True)
    ingredients = models.TextField(blank=True)
    nutrition = models.JSONField(default=dict, blank=True)
    is_active = models.BooleanField(default=True)
    is_sponsored = models.BooleanField(default=False)
    sponsored_until = models.DateTimeField(null=True, blank=True)
    total_sales = models.PositiveIntegerField(default=0)
    view_count = models.PositiveIntegerField(default=0)
    cart_add_count = models.PositiveIntegerField(default=0)
```

15.3. Delivery model

```
class DeliveryTask(models.Model):
    STATUS = [
        ('unassigned', 'Unassigned'), ('offered', 'Offered'), ('accepted', 'Accepted
        ('picked_up', 'Picked up'), ('delivered', 'Delivered'), ('failed', 'Failed')
    ]
    order = models.OneToOneField('orders.Order', on_delete=models.CASCADE)
    courier = models.ForeignKey('couriers.Courier', null=True, blank=True, on_delete=
    status = models.CharField(max_length=20, choices=STATUS, default='unassigned')
    pickup_latitude = models.FloatField()
    pickup_longitude = models.FloatField()
    dropoff_latitude = models.FloatField()
    dropoff_longitude = models.FloatField()
    estimated_distance_m = models.PositiveIntegerField(default=0)
    estimated_duration_s = models.PositiveIntegerField(default=0)
    assigned_at = models.DateTimeField(null=True, blank=True)
    picked_up_at = models.DateTimeField(null=True, blank=True)
    delivered_at = models.DateTimeField(null=True, blank=True)
```

16. API dizayni

16.1. Auth

```
POST /api/v1/auth/request-otp/
POST /api/v1/auth/verify-otp/
POST /api/v1/auth/refresh/
POST /api/v1/auth/logout/
GET  /api/v1/me/
PATCH /api/v1/me/
```

16.2. Catalog

```
GET  /api/v1/products/
GET  /api/v1/products/{slug}/
GET  /api/v1/products/{id}/ar/
GET  /api/v1/categories/
GET  /api/v1/shops/
GET  /api/v1/search/?q=&lat=&lng=&lang=
```

16.3. Cart va order

```
GET    /api/v1/cart/
POST   /api/v1/cart/items/
PATCH  /api/v1/cart/items/{id}/
DELETE /api/v1/cart/items/{id}/
POST   /api/v1/checkout/quote/
POST   /api/v1/orders/
GET    /api/v1/orders/
GET    /api/v1/orders/{id}/tracking/
POST   /api/v1/orders/{id}/repeat/
```

16.4. Payment

```
POST /api/v1/payments/create/
POST /api/v1/payments/click/callback/
POST /api/v1/payments/payme/callback/
GET  /api/v1/payments/{id}/status/
POST /api/v1/payments/{id}/refund-request/
```

16.5. AI

```
POST /api/v1/ai/chat/messages/
POST /api/v1/ai/design/generations/
GET  /api/v1/ai/design/generations/{id}/
POST /api/v1/ai/products/{id}/copy/
POST /api/v1/ai/products/{id}/translate/
```

16.6. AR

```
POST /api/v1/seller/products/{id}/ar/capture/
GET  /api/v1/seller/products/{id}/ar/jobs/{job_id}/
GET  /ar/{product_id}/
GET  /api/v1/products/{id}/ar/qr/
```

17. Background jobs

Celery queues:

```
critical: payment callbacks/reconciliation
high: order notifications/courier dispatch
normal: AI chat async analytics, image optimization
low: AI poster, translations, AR/3D conversion, reports
```

Vazifalar:

- `send_order_notifications`
- `assign_courier`
- `recalculate_eta`
- `sync_payment_status`
- `generate_ai_poster`
- `generate_product_copy`
- `translate_product`
- `start_3d_capture`
- `poll_3d_job`
- `generate_ar_qr`
- `send_crm_campaign`
- `daily_seller_report`
- `cleanup_expired_carts`

18. Xavfsizlik va compliance

18.1. Secretlar

```
DJANGO_SECRET_KEY=
DATABASE_URL=
REDIS_URL=
TELEGRAM_BOT_TOKEN=
CLICK_MERCHANT_ID=
CLICK_SERVICE_ID=
CLICK_API_KEY=
PAYME_MERCHANT_ID=
PAYME_KEY=
AI_PROVIDER=
AI_API_KEY=
S3_ACCESS_KEY=
S3_SECRET_KEY=
```

- `.env` gitga kirmaydi.

- `.env.example` faqat key nomlari bilan bo‘ladi.
- Oldin oshkor bo‘lgan Telegram token rotatsiya qilinadi.
- Production key rotation jadvali bo‘ladi.

18.2. Web security

- HTTPS only
- CSP, HSTS, secure cookies
- CSRF protection
- Rate limiting: OTP, login, chat, checkout
- Input validation va file upload scan
- RBAC va object-level permissions
- Audit log: payment, refund, role, price, stock, order status
- Backup restore drills

18.3. AI security

- Prompt injectionga qarshi tool allowlist
- Chatbot system promptda secret/DB dump bermaydi
- PII modelga minimal yuboriladi
- Kredit card data modelga yuborilmaydi
- User-generated content moderation
- AI outputni actionable commandga aylantirishdan avval server validation

19. Monitoring va analitika

19.1. Texnik monitoring

- Error rate, latency, 5xx
- Queue depth va failed jobs
- DB slow query
- Payment callback failures
- AI cost per feature/shop
- CDN bandwidth
- Uptime checks

19.2. Business KPI

- GMV
- Net revenue/commission
- Orders/day
- Conversion rate
- Average order value
- Repeat purchase rate
- Delivery SLA
- Cancellation/refund
- Customer acquisition cost
- LTV/CAC
- Seller active rate
- AI poster to conversion uplift
- AR view to add-to-cart uplift

20. UI/UX qoidalari

- Mobile-first; Guliston foydalanuvchilarining internet tezligini hisobga oling.
- First content < 2.5 s maqsad.
- Rasm: WebP/AVIF, responsive sizes, lazy load.
- Skeleton loading.
- PWA install banner.
- Accessibility: katta touch target, kontrast, form validation.
- Checkout 3 ekrandan oshmasin.
- Har buyurtmada aniq: summa, delivery, ETA, shop, payment status.
- “AI generated” dizayn sellerga koʻrsatilishi va edit qilinishi kerak.

21. Flutter ilova

21.1. Modullar

- Auth va OTP
- Home/catalog/search
- Product + AR deep link
- Cart/checkout/payment WebView
- Map/order tracking

- Chatbot
- Push notification
- Offline catalog/cache
- In-app browser allowlist

21.2. Tavsiya stack

```
Flutter
Dio
Riverpod yoki Bloc
flutter_secure_storage
Firebase Messaging
flutter_map yoki native map SDK
WebView/InAppWebView payment uchun
model-viewer WebView yoki platform-native AR deep link
Hive/Isar offline cache
```

21.3. Deep links

```
guliston://product/{slug}
guliston://order/{id}
guliston://ar/{id}
guliston://payment/result/{id}
https://domain.uz/ar/{id}
```

22. Delivery bosqichlari

Phase 0 — Security and foundation (1–2 hafta)

- Telegram token rotate
- Env migration
- PostgreSQL/Redis/Celery
- Docker, CI/CD, backup
- Permission audit
- Payment provider interface + mock provider

Phase 1 — Commerce reliability (3–5 hafta)

- Order state machine
- Stock reservation
- Payment mock end-to-end
- Courier dispatch basic

- Delivery zones/fees
- Production monitoring
- Checkout UX

Phase 2 — Payment and logistics (3–4 hafta)

- Click sandbox/production integration
- Payme sandbox/production integration
- Webhook verification/reconciliation
- 2–3 in-house courier pilot
- Aggregator/Yandex feasibility and adapter
- OTP delivery confirmation

Phase 3 — AI chatbot MVP (3–4 hafta)

- FAQ + intent router
- Product search tool
- Cart/order/tracking tools
- UZ/RU/EN
- Chat history and feedback
- Operator handoff
- Cost dashboard

Phase 4 — Seller AI (3–4 hafta)

- AI product copy
- Translation
- Poster MVP
- Pro plan credits
- Moderation
- Analytics

Phase 5 — AR/3D MVP (3–5 hafta)

- Manual GLB/USDZ upload
- Model-viewer product page
- QR code
- 10 premium SKU pilot
- 3D capture provider integration
- Size verification workflow

Phase 6 — Flutter and scale (6–10 hafta)

- Customer app MVP
- Courier app/PWA
- Push
- In-app browser
- Offline cache
- Performance test

23. 50–100 mln so‘m budjet strategiyasi

50–100 mln so‘m bilan hamma funksiyani bir vaqtda ideal qilish real emas. Eng yaxshi yo‘l — asosiy savdo va yetkazishni daromad beradigan darajaga olib chiqib, AI/AR’ni pilot sifatida qo‘shish.

23.1. 50 mln so‘m ssenariy

Yo‘nalish	Taxminiy budjet	Izoh
Core backend/production hardening	10–15 mln	mavjud kodni ishonchli qilish
Payment module + test/sandbox	5–8 mln	API kalit kutiladi
Courier/delivery MVP	8–12 mln	1–2 kuryer/pilot
AI chatbot MVP	6–10 mln	FAQ + tools, minimal UI
AI seller copy/poster pilot	4–7 mln	kreditli limit
Marketing/launch	8–12 mln	lokal fokus
Infra/API reserve	3–5 mln	server, SMS, AI, storage

Bu ssenariyda AR faqat manual 3D upload va 5–10 SKU demo bo‘ladi.

23.2. 100 mln so‘m ssenariy

Yo‘nalish	Taxminiy budjet	Izoh
Production core va QA	15–20 mln	test/monitoring/security
Payment + finance operations	7–10 mln	provider integration
Logistics pilot	15–25 mln	2–3 kuryer, transport/operatsiya
AI chatbot + seller AI	15–20 mln	multilingual + analytics
AR/3D pilot	8–15 mln	capture, QA, 20–50 SKU
Flutter MVP	15–25 mln	customer app first
Marketing + reserve	10–15 mln	launch/experiments

23.3. Budjetning asosiy qoidasi

- Avval GMV, repeat rate va delivery SLA’ni isbotlang.
- Marketing xarajatini tracking links, promo code va cohort bilan o‘lchang.
- AI va AR uchun kredit/limit bo‘lsin; “cheksiz AI” modeli zarar keltiradi.
- Yandexga ulanishdan oldin o‘z delivery cost/order va SLA’ni o‘lchang.

24. 90 kunlik ish rejasi

1–30 kun

- Security cleanup va token rotation
- PostgreSQL/Redis/Celery
- Payment abstraction + mock checkout
- Delivery pricing/zones
- Courier dashboard improvements
- Seller permissions
- Analytics event schema

31–60 kun

- Click/Payme onboarding, test integration
- Order/payment reconciliation
- 2 kuryerli lokal pilot
- AI chatbot FAQ + product search
- AI product copy/translation
- CRM: abandoned cart, reorder

61–90 kun

- AI poster Pro beta
- Model-viewer va QR AR pilot
- 10–20 mahsulot 3D demo
- Referral/loyalty
- A/B tests
- Flutter MVP scope freeze/start

25. Test rejasi

25.1. Payment tests

- Success callback
- Failed callback
- Duplicate callback
- Wrong amount
- Invalid signature
- Timeout
- Refund
- Concurrent checkout

25.2. Delivery tests

- Zone ichida/tashqarisida address
- Courier accept/reject/timeout
- GPS unavailable
- Order cancellation before/after pickup
- OTP mismatch

25.3. AI tests

- Uzbek/Russian/English intent
- Prompt injection
- Tool permission boundary
- AI hallucinated price
- Cache hit/miss
- Credit limit
- Image moderation

25.4. AR tests

- Android Chrome WebXR
- Android fallback
- iOS Safari Quick Look
- Unsupported browser fallback
- Large model slow network
- Correct physical scale

26. Qabul mezonlari

Commerce

- Checkout 99% successful test flow
- Payment callback idempotent
- Stock oversell yo‘q
- Order status history saqlanadi

Delivery

- Courier assignment 60 soniyadan kam
- Customer tracking ishlaydi
- Delivery fee quote server-side aniqlanadi

AI

- FAQ savollarining kamida 70% AI chaqirmasdan javob oladi
- Product search natural language orqali ishlaydi
- Har tool action permission/audit bilan
- Monthly AI budget limit mavjud

AR

- Kamida 10 SKU uchun GLB preview
- Kamida Android/iOS fallback test
- QR URL ishlaydi
- O‘lcham metadata ko‘rsatiladi

Language

- UZ/RU/EN UI 100% asosiy flowlarda
- Checkout va payment yuridik matnlari tasdiqlangan tarjimada
- Chatbot barcha 3 tilda ishlaydi

27. Dasturchi uchun final prompt

Quyidagi promptni AI coding agent yoki dasturchi jamoaga bering:

```
Sen Online Store Guliston loyihasining senior Django/DRF, Flutter, DevOps va AI integ

Muhim qoidalar:
1. Mavjud ishlaydigan customer/admin/courier/map/telegram funksiyalarini buzma.
```

```
2. Har o‘zgarishdan oldin kodbazani tekshir, keyin kichik commitga bo‘lingan impleme
3. SQLite MVP bo‘lsa ham production uchun PostgreSQL, Redis, Celery, S3 storage, Doc
4. Hech qachon secret, bot token, API keyni kodga yozma; `.env.example` yarat, `.env
5. Oldin oshkor qilingan Telegram tokenni unsafe deb hisobla; yangi token berilmagun
6. To‘lov API kalitlari hali yo‘q. PaymentProvider adapter, MockPaymentProvider, Ord
7. Payment callbackda signature, amount, order, replay/idempotency tekshiruvini majbu
8. AI chatbot tool-based bo‘lsin: product search, cart, order tracking, seller analy
9. Chatbot state-changing actionlarida preview + explicit confirmation + role permiss
10. UZ/RU/EN barcha customer, seller, courier va chatbot flowlarida ishlasin.
11. AI xarajatlarini kamaytir: FAQ/rule layer, cache, intent routing, qisqa JSON outp
12. AI poster funksiyasini background Celery jobda, moderatsiya va seller credit bila
13. AR’ni avval manual GLB/USDZ upload + model-viewer + QR page bilan qil. Keyin 3D
14. Qidiruv ranking organik quality/relevance/stock/SLA/distancega asoslanadi. Spons
15. Logistics gibrid bo‘lsin: own courier dispatch asosiy, external/Yandex adapter fa
16. Har yangi endpoint uchun serializer validation, permissions, tests va OpenAPI do
17. Har feature yakunida migration, unit test, integration test, qisqa changelog va
18. Mobile-first responsive templatesni saqla; Flutter API v1 contractni stable qil.

Implementatsiya tartibi:
A. Security/infra: env, PostgreSQL config, Redis/Celery, Docker, logs, backups.
B. Commerce reliability: inventory, order state machine, payment abstraction/mock, a
C. Delivery: zones, quote, courier assignment, tracking, OTP proof.
D. Click/Payme adapters: sandbox-ready, webhook verification, reconciliation.
E. AI chatbot MVP: FAQ, multilingual intent, tools, human handoff, cost control.
F. Seller AI: product copy, translation, poster generation, Pro credits.
G. AR MVP: GLB/USDZ, model-viewer, QR, dimensions, job model.
H. Analytics/CRM: events, referral, loyalty, campaigns.
I. Flutter API: JWT/OTP, catalog/cart/order/payment/tracking/chat endpoints.

Har bosqichda avval reja, keyin faylma-fayl o‘zgarishlar, so‘ng test buyruqlari va e
```

28. Yakuniy prioritet

1. **Xavfsizlik:** tokenlar, env, backup, permission.
2. **Pul oqimi:** checkout, payment adapter, order state machine.
3. **Operatsiya:** yetkazish, kuryer, SLA, support.
4. **O‘sish:** CRM, referral, reviews, katalog sifati.
5. **AI:** avval chatbot/SEO/copy; keyin poster.
6. **AR:** avval premium kategoriyalarda pilot, keyin scale.
7. **Mobil ilova:** API stable bo‘lgandan keyin Flutter.

Bu yo‘l bilan loyiha shunchaki funksiyalar yig‘indisi emas, balki ishonchli va iqtisodiy jihatdan boshqariladigan lokal e-commerce ekotizimiga aylanadi.